LECTURE 25

24 November 2023 09:35

$$f(x,y) = x^2y + x + y + 5$$

$$f_x = 2xy + 1 \qquad f_y = x^2 + 1$$

$$\int 2xy + 1 \, dx = yx^2 + x + \phi(y) \qquad \int x^2 + 1 \, dy = x^2y + y + \psi(x)$$

$$\int_0^x x^2y + x + y + 5 \, dy = \left[\frac{x^2y^2}{2} + xy + \frac{y^2}{2} + 5y\right]_0^x$$

$$= \frac{x^4}{2} + x^2 + \frac{x^2}{2} + 5x$$

$$= \frac{x^4}{2} + \frac{3x^2}{2} + 5x$$

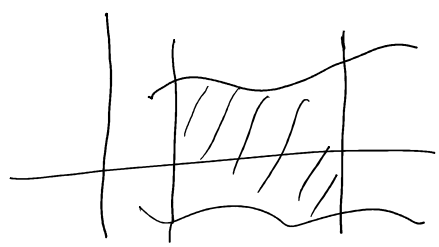
$$\int_1^2 \left(\int_0^x x^2y + x + y + 5 \, dy\right) dx = \int_1^2 \frac{x^4}{2} + \frac{3x^2}{2} + 5x \, dx$$

Multiple integration or Iterated integration $\qquad$ H.W. = 14.1

$$\int_a^b \int_{f_1(y)}^{f_2(y)} f(x,y) \, dx \, dy \qquad \int_c^d \int_{h_1(x)}^{h_2(x)} f(x,y) \, dy \, dx$$

Geometrical interpretation:

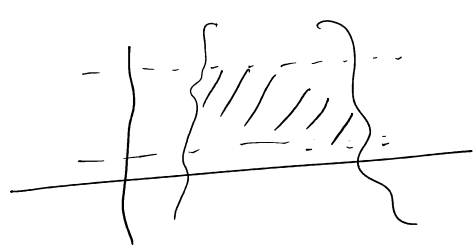
$$\int_a^b \int_{f_1(x)}^{f_2(x)} dy \, dx$$



$x = a \qquad y = f_2(x)$

$x = b \qquad y = f_1(x)$

$$\int_c^d \int_{h_1(y)}^{h_2(y)} 1 \, dx \, dy$$



① $\int^2 \int_1^{2x} dy \, dx = \underline{\text{change}}$ of $\qquad y = x^2 \qquad y = 2x$

① $\int_0^2 \int_{x^2}^{2x} 1 \, dy \, dx$ = change of order of integration

$y = x^2 \quad y = 2x$

$= \int_0^2 (y)_{x^2}^{2x} \, dx$

$= \int_0^2 (2x - x^2) \, dx$

$= \frac{4}{3}$

2) $\int_0^4 \int_{y/2}^{\sqrt{y}} 1 \cdot dx \, dy$

$\int_0^4 (x)_{y/2}^{\sqrt{y}} \, dy$

$\int_0^4 \left(\sqrt{y} - \frac{y}{2}\right) dy$

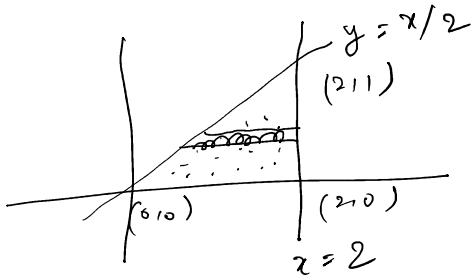
$= 4/3$

Note: If limits are just constants. then by changing the o.o.i. value of integral does'nt change.

3) $\int_0^2 \int_0^{x/2} 1 \cdot dy \, dx$

$y = 0 \quad y = x/2$

$x = 0 \quad x = 2$



$\int_0^1 \int_2^{2y} dx \, dy$

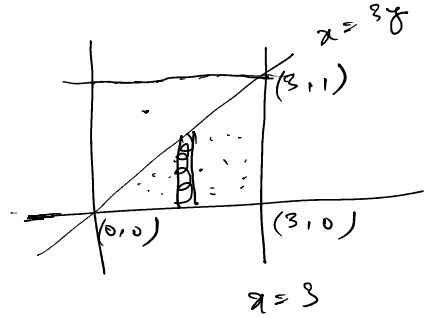
H.W. 1) $\int_0^1 \int_0^{2x} dy \, dx$

Ans: $\int_0^2 \int_{y/2}^1 dx \, dy$

4) $\int_0^1 \int_{3y}^3 e^{x^2} \, dx \, dy$

$x = 3y \quad x = 3$

$y = 0 \quad y = 1$



$\int_0^3 \int_0^{x/3} e^{x^2} \, dy \, dx$

$= \int_0^3 e^{x^2} \left(\int_0^{x/3} 1 \, dy \right) dx$

$= \int_0^3 e^{x^2} (y)_0^{x/3} \, dx$

Ans: $\int_0 \quad)_{y/2}$

$$= \int_0^3 e^{x^2} \cdot \frac{x}{3} \, dx$$

$x^2 = t$

$$= \frac{1}{3} \int_0^9 e^t \, \frac{dt}{2}$$

$2x \, dx = dt$